

Roll No. _____

25/1192**B.C.A. (Fifth Semester)****Examination, 2025****Third Paper****(Computer Networks)***Time : Two Hours / [Maximum Marks : 75***Note :** Attempt **all** sections as per instructions.**Section-A****(Very Short Answer Type Questions)****Note :** Attempt all the **five (05)** questions.

Each question carries **two (02)** marks and the answer of each question should not exceed 50 words. $5 \times 2 = 10$

1. (a) Define Distributed processing. What are its advantages?
- (b) Name the different network layers in the OSI model and show their hierarchy with the help of a diagram.

P.T.O.**25/1192**

- (c) What is the function of a Modem in a Network communication scheme?
- (d) Differentiate between Guided and Unguided media with diagram.
- (e) Explain what is meant by Authentication, Error control.

Section-B**(Short Answer Type Questions)**

Note : Attempt any the **five (05)** questions out of 08 (eight) questions. Each question carries 05 (five) marks and the answer of each question should not exceed 100 words. $5 \times 5 = 25$

2. (a) Differentiate between Time Division Multiplexing and Frequency Division Multiplexing, stating the relative merits of each scheme.
- (b) Define and explain:
 - (i) Attenuation
 - (ii) Shannon Capacity
 - (iii) Propagation Speed
 with respect to transmission Media.

25/1192

- (c) State and explain the purpose of the following layers:
- (i) Data Link Layer
 - (ii) Network layer
- (d) What do you understand by
- (i) Gateway
 - (ii) Bridge in a WAN
- (e) What are the types of Topologies that may be used to implement a network? Explain briefly.
- (f) Differentiate between circuit switching packet switching and message switching indicating the advantage of each of these methods.
- (g) What are the functions of the transport layer? Discuss briefly.
- (h) Explain ISDN and its services.

Section-C

(Long Answer Type Questions)

Note : Attempt any 02 **(two)** questions out of total 04 **(four)** questions. Each question carries 20 **(twenty)** marks and answer of each question should not exceed **400** words. $2 \times 20 = 40$

25/1192

3. (a) Write an explanatory note on the following: 6+8+6
- (i) Categories of Networks
 - (ii) Transmission Media
 - (iii) PPP layers
- (b) Differentiate between the following with examples: 7+7+6
- (i) Connection Oriented Protocols and connectionless Protocols.
 - (ii) Repeaters and Routers
 - (iii) Flow control and Congestion control.
- (c) What is the organization of the TCP/IP model? Discuss the function of each layer with a neat Diagram.
- (d) Write the function of the session, presentation and Application layers in the OSI model. Also mention the Network Services that are inherently based in each of the above layers.